9.1 – Solutions and Molarity

1. A solution is a _____ mixture that consists of two parts. Write out the parts below and define each.

    a. _____:

    b. _____:

2. Write an equation for each of the following substances below when they dissolve in water.

    a. $NaCl\ (s)$ □

    b. $CH_3OH\ (l)$ □

    c. $MgCl_2\ (s)$ □

    d. $Mg(OH)_2$ □

3. We can quantitatively describe a solution using molarity. What needs to be known about the solution to determine the molarity?

    a. molarity:

4. A student prepares a potassium fluoride solution by dissolving 1.00 moles of potassium fluoride to make 0.100 liters of solution. What is the molarity of that solution?

5. A student dissolves 15.0 grams of solid magnesium nitrate to make 0.250 liters of solution. Calculate the molarity of that solution.

6. How many grams of calcium chloride are present in 150.0 mL of a 1.25 M solution of calcium chloride?

7. How many liters of NaCl solution can be prepared with a concentration of 0.750 molar using 1.50 moles of NaCl?

8. An aqueous solution is prepared by diluting 3.0 mL of acetone(C_3H_6O) (density = 0.789 g/mL) with water to a final volume of 75.0 mL. The density of water is 1.00 g/mL.

    a. Calculate the molarity of the solution above.

9.2 – Solution Creation

1. A student wishes to perform an experiment with a solution of aluminum nitrate. The student needs to prepare 500.0 mL of 0.750 M aluminum nitrate. The student only has access to 5.00 M aluminum nitrate.

a. Calculate the number of mL of 5.00 M aluminum nitrate that needs to be diluted to prepare 500.0 mL of 0.750 M aluminum nitrate.

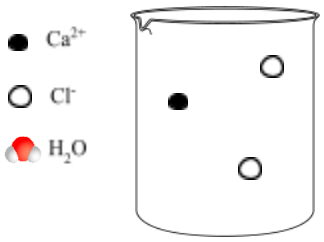
b. Discuss the steps that the student should take to prepare this solution.

c. The aqueous aluminum nitrate reacts with sodium hydroxide in a double replacement reaction. Write the balanced chemical equation for this reaction.

2. A student prepares a solution of calcium chloride by dissolving 25.0 grams of calcium chloride to make 750.0 mL of solution.

a. Write the equation for the dissociation of calcium chloride.

b. In the beaker provided, show how the water molecules interact with the ions in solution. Consider the charges of each and how they will arrange themselves. Draw at least 2 water molecules around each ion.



c. Calculate the molarity of the solution created above.

d. The solution is left outside in the sun and 150.0 mL of water evaporates from the solution. Calculate the new molarity of the solution.

3. A student needs to prepare 500.0 mL of 1.77 M nitric acid.

a. The formula for nitric acid is: _____

b. What part of the formula makes this substance an acid? _____

c. The student has access to a 10.0 M stock solution of nitric acid. Determine how many mL of 10.0 M needs to be diluted to prepare 500.0 mL of 1.77 M nitric acid

d. Discuss the steps the student should take to prepare this solution.

9.3 – Solution Stoichiometry

1. Aqueous Silver nitrate reacts with aqueous potassium chloride to form aqueous potassium nitrate and solid silver chloride.
   a. Write the balanced chemical equation for the reaction.

   b. What mass of silver chloride will form if 50.0 mL of 0.225 M silver nitrate reacts with excess potassium chloride?

   c. How many moles of potassium chloride will react with 50.0 mL of 0.225 M silver nitrate?

   d. Using your answer from part C, determine what volume, in mL, of potassium chloride reacted if the concentration is 0.500 M.

2. Aqueous barium chloride reacts with aqueous aluminum sulfate to produce solid barium sulfate and aqueous aluminum chloride.
   a. Write the balanced chemical equation for the reaction.

   b. What mass of solid barium sulfate will form if 15.0 mL of 0.750 M aluminum sulfate reacts with XS barium chloride.

   c. How many moles of barium chloride reacted with the 15.0 mL of 0.750 M aluminum sulfate?

   d. Using your answer from part C, determine what volume, in mL, of barium chloride reacted if the concentration used was 0.225 M.

3. Aqueous hydrochloric acid reacts with sodium hydroxide to form water and aqueous sodium chloride.
   a. Write the balanced chemical equation for the reaction.

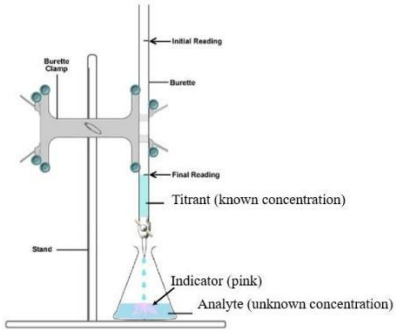
   b. What volume, in mL, of 0.150 M HCl will be needed to react with 35.0 mL of 0.325 M sodium hydroxide?

9.4 – Titrations

1. Describe the purpose of a titration:

2. Why are indicators necessary for titrations? What is the purpose of an indicator?

2. A solution of HCl with an unknown concentration is analyzed. A titration will be performed to determine the concentration of the acid. The diagram provided below shows the student's experimental setup. The data table provides all necessary information to calculate the concentration of the unknown acid.



Chemical Reaction: $HCl + NaOH \rightarrow H_2O + NaCl$

Volume of Acid	25.0 mL
Conc. of NaOH	0.650 M
mL of base to reach end point	12.75 mL

a. The student performs the titration and the data above are obtained. The end point is reached after 12.75 mL of the NaOH solution is added.

i. Calculate the number of moles of HCl that was present in the original unknown acid solution.

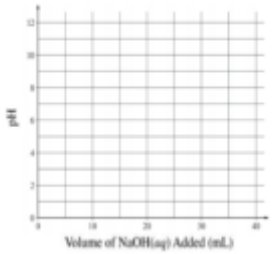
ii. Calculate the concentration of the HCl solution.

4. A student performs a titration with acetic acid ($HC_2H_3O_2$) and sodium hydroxide (NaOH)

a. Write a balanced chemical equation for this reaction.

The student titrations 10.0 mL of the acetic acid solution with an unknown concentration. The student uses 0.100 M NaOH to perform the titration. The pH of the solution is monitored, and the following data is obtained.

a. How many mL of the sodium hydroxide solution were required to reach the end point? _____



b. Calculate the number of moles of sodium hydroxide used to reach the end point.

c. Calculate the original concentration of the acetic acid solution.